

THE PHYSICS BEHIND RAINBOWS



Scientists have been trying to understand the unusual physics behind rainbows for a very long time now. What is our current understanding on this meteorological phenomenon? Here is an interesting research on the science behind the rainbow.
<http://dgit.in/PRainbow>

SENDING SCIENTISTS TO WAR



"Mental trauma is more complicated than we think" say combat veterans. "To reduce psychological casualties we need researchers on the battlefield. To truly understand the symptoms of Post Traumatic Stress Disorder, we need scientific boots on the ground."
<http://dgit.in/ScientistW>

SCIENCE IS ALL ABOUT BUILDING MODELS



Rhett Allain, a professor of physics, explains why one shouldn't dare teach science without building relevant models. A model can be a mathematical equation, a physical model, or even a conceptual model such as the photosynthesis cycle.
<http://dgit.in/ScModels>

3D PRINTING A 'GHOST GUN'



'Ghost Gun' is a popular term adopted by gun control advocates and gun lovers as well, for untraceable guns with no serial number. And someone actually 3D printed an untraceable, fully functional, accurate AR-15 semi-automatic rifle.
<http://dgit.in/AR153D>

of these three systems essentially work in tandem. The natural language processing engine was custom-built by using basic keywords, and then building on them. For instance, using the words "bedroom", "lights" and "dim" gave a specific command to Jarvis - dim the lights of the bedroom. However, when two people in the same house are using the same assistant from the same interface to give similar commands, things could get more complicated since Jarvis would not know which bedroom is being specified.

This is where the voice engine and its recognition features kicked in. Zuckerberg, in his note, stresses heavily on the importance of context when it comes to an AI, and rightly so. It took more hours to teach his AI this than any other aspect. As a result, and with more commands given to it, Jarvis gradually picked up "context" in a human way - understanding who is giving a particular instruction and responding to that very person's own usage and preference history, rather than relying on generic commands.

This is one aspect that Zuckerberg particularly elucidated on by using music. For instance, "play something light" for him certainly meant something different in comparison to the same command given

by his wife. Music is also trickier to be taught to an AI assistant, as unlike household chores, creative topics will always have more subjective responses. For this, Jarvis was taught to look at Zuckerberg's past listening history, most played tracks, and more to gauge the kind of music that he would like. The same goes for the other members of his house.

Facial recognition was used by Zuckerberg in order to recognise those that visited his house. He fit multiple cameras on his doorway and connected them to his home network to give Jarvis a live feed. When anyone would reach his place, the multiple cameras would show various angles, and as a result, make it easier to get a direct view of the person at the door. This further allowed Jarvis to better "recognise" him or her by tallying with photographs and facial patterns of Zuckerberg's trusted contacts.

While all of this, on paper, sounds fairly straightforward, it really isn't. The trick is to get all the relevant machines of a house to connect to the Internet. While the Internet of Things is on a definite rise, most of our household elements are still not connected to the Internet. Even the ones that still have individual protocols and mechanisms, which means that each speaks a different language. This is the more difficult bit of the entire task, than getting the appliances to connect to the Internet.

Jarvis is an AI-powered assistant that became a common platform - a standard protocol of sorts for everything that's connected. This is the crucial bit of Zuckerberg's experiment, as this not only exhibited the limitations of the present realm of IoT, but also stressed on the need for a common platform for smart households and IoT to really flourish. We may have

individual, connected multi-speaker units across multiple rooms, but they may still not recognise the voice and AI interface of your smartphone, which in all possibility will use a different interface. As a result, this would still require you to use different interfaces to access different appliances at home, completely belying the point of a connected home



However, it is crucial to note that this is not exactly new. Amazon Echo and Google Home were made to be similar smart home hubs for all things connected. While they do not (yet) have the power to control every single appliance that you would use at home, these are very similar foundations that have been laid down. Jarvis is not a new invention per se, but is more of a refinement that works on creating a single platform for everything, without the need for different apps to control each.

It is this very aspect that Mark Zuckerberg has addressed, and this is what Jarvis does for a living - connect the entire house, understand personal contexts, and somewhat recognise you to carry out personal tasks. Jarvis is, in a fascinatingly simplistic way, the forerunner to personal home AI assistants that our future will hold.

Oh, and he's also Morgan Freeman. 

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Music is understandably trickier for bots and AI